Tripurana Sai Shashank

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About Me

Electronic Engineer with a strong foundation in embedded systems, IoT, machine learning, and VLSI design. Skilled in RTL design, Verilog, and finite state machines for efficient digital logic implementation. Driven to develop intelligent, hardware-optimized solutions for practical and impactful AI applications.

Education

Gayatri Vidya Parishad College of Engineering, Btech in ECE — Expected grad.2026

- CGPA: 8.72/10

Sri Chaitanya colloge, MPC — 2020-2022

- Percentage:90.4

TECHNICAL STRENGTHS

Languages: C,Python,Verilog .
Tools: Cadence,esi m,Matlab,Xilinx,HFSS
Technologies: TensorFlow,OpenCV
Webframeworks: Flask,FastAPI

EXPERIENCE

Research Intern

University of Calcutta | Apr 2025 – Jul 2025

- Developed **Autonomous Garden**, a smart plant-care system using ESP32, sensors, InfluxDB, and a fine-tuned LLM (Lit-GPT) for real-time environmental monitoring

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- Led **H.A.R.V.E.S.T.**, applying transformer models to hydrogel-soil interaction data to compute Salt Responsive Index and optimize nutrient retention..
- Gained interdisciplinary experience across IoT, ML, DL, and LLM-based modeling for sustainable agriculture applications.

Projects

Smart Traffic light System for Ememrgency Vehciles

Technologies/Hardware: *Arduino Uno, RF Module, Arduino IDE*

- Developed a smart 4-way traffic light system using Arduino Uno that detects emergency vehicles and dynamically adjusts signals to prioritize their passage, improving response times and traffic flow.

American Sign Language Detection

Technologies: *python,OpenCV,Tensorflow,Flask*

- Developed a machine learning model to accurately detect American Sign Language (ASL) gestures, using a custom dataset for training to enhance model precision and reliability

VLSI-GPT

Technologies: *Python,LLM,Flask,TTS.*

- Built a web-based VLSI Query Portal with real-time text and voice input, delivering AI-powered responses in both audio and text. Used Flask for backend and Web Speech API for interactive voice features.

Talking Groot

Technologies/Hardware: *ESP-8266,Google FireBase-Realtime firebase,python,LLM,TTS*

- A fun and interactive project where a plant, represented by "Groot," responds and "talks" based on real-time environmental sensor data (like temperature, humidity,). This project brings plants to life by giving them a personality and voice using smart technology.

FIFO (First-In-First-Out)

Technologies: *Verilog,Xilinx*

- Designed a synchronous FIFO buffer with configurable data width and depth, featuring read/write pointers, full and empty status flags, and reliable data transfer between producer and consumer modules.

Student Leadership Roles

Documentation Lead: WeR4Help,Gvpce(A)

Marketing Lead: AI BUILD CLUB(ABC),Gvpce(A)

excom board member: IEEE,Gvpce(A)

President: AI BUILD CLUB(ABC),Gvpce(A)